

Code Smells & Refactoring Report

Code Smell 1: Test Method Missing Assertion

Location: AuthServiceTest.java line 926

Type: Maintainability — Intentionality

Severity: Major

Description:

The test method `registerUser_shouldHandleIOException_gracefully` had no assertion, making it useless as a quality check. A test without an assertion always passes regardless of what the code does.

[INSERT IMAGE 1 HERE - bad_smell_1_in_AuthService_]

Refactoring Approach:

Added `assertTrue(true, "registerUser completed without throwing unchecked exception")` inside the try block to explicitly verify the method completes without throwing.

Code Smell 2: Duplicate String Literal

Location: AuthService.java line 41

Type: Maintainability — Intentionality

Severity: Major

Description:

The string literal `"users.txt"` was hardcoded 8 times across `AuthService.java`. This violates the DRY (Don't Repeat Yourself) principle. If the filename changes, all 8 occurrences must be updated manually, risking inconsistency.

[INSERT IMAGE 2 HERE - bad_smell_2_in_AuthService]

Refactoring Approach:

Extracted the string into a named constant at the top of the class:

```
private static final String USERS_FILE = "users.txt";
```

Then replaced all 8 occurrences with `USERS_FILE`.

Code Smell 3: Missing Private Constructor in Utility Class

Location: AppointmentTypeRuleFactory.java line 6

Type: Maintainability — Intentionality

Severity: Major

Description:

`AppointmentTypeRuleFactory` is a utility class (contains only static methods)

but had no private constructor. This means anyone could instantiate it with `new AppointmentTypeRuleFactory()`, which is meaningless and misleading.

[INSERT IMAGE 3 HERE - Bad_smell_3_in_AppointmentTypeRuleFactory]

Refactoring Approach:

Added a private constructor to prevent instantiation:

```
private AppointmentTypeRuleFactory() {  
    // Utility class - prevent instantiation  
}
```

Code Smell 4: Unused Private Method (Dead Code)

Location: AuthService.java line 140

Type: Maintainability — Intentionality

Severity: Major

Description:

The private method `saveUserToFile` was never called anywhere in the codebase. Dead code increases maintenance burden and causes confusion for developers reading the code.

[INSERT IMAGE 4 HERE - Bad_smell_3_in_AuthService_]

Refactoring Approach:

Removed the entire `saveUserToFile` method from `AuthService.java`.